

3.

$$a) f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n)) \wedge f(n) = \Omega(g(n))$$

$$f(n) = \Theta(g(n)) \quad c_1, c_2 > 0$$

$$n_0 \quad 0 \leq c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$$

$$n \geq 0$$

$$f(n) = O(g(n)) \text{ donde } c = c_2 \quad \leftarrow$$

$$f(n) = \Omega(g(n)) \text{ donde } c = c_1 \quad \leftarrow$$

$$b) \max(f(n), g(n)) = \Theta(f(n) + g(n))$$

$$n_0 \text{ tal que } f(n) \geq 0 \text{ y } g(n) \geq 0 \quad \forall n \geq n_0$$

$$c_2 = 1 \quad (\text{Cota superior})$$

$$h(n) = f(n) \vee g(n) \quad \rightarrow h(n) \leq f(n) + g(n)$$

$$c_1 = \frac{1}{2} \quad (\text{Cota inferior})$$

$$f(n) \leq h(n) \text{ y } g(n) \leq h(n)$$

$$\downarrow$$

$$f(n) + g(n) \leq 2h(n) \quad \rightarrow \frac{1}{2}(f(n) + g(n)) \leq h(n)$$

$$\forall n \geq n_0:$$

$$0 \leq \frac{1}{2} \cdot (f(n) + g(n)) \leq \max(f(n), g(n)) \leq 1 \cdot (f(n) + g(n))$$

$$c) f(n) = O(g(n)) \wedge g(n) = O(h(n)) \rightarrow h(n) = O(h(n))$$

$$= h(n)$$

$$= \Omega(f(n))$$

Suposiciones

$$\left. \begin{array}{l} f(n) = O(g(n)) \\ g(n) = O(h(n)) \end{array} \right\} \begin{array}{l} \text{Existen } c_1 > 0, 0 \leq f(n) \leq c_1 \cdot g(n) \quad \forall n \geq n_1 \\ \text{Existen } c_2 > 0, 0 \leq g(n) \leq c_2 \cdot h(n) \quad \forall n \geq n_2 \end{array}$$

$$\text{Si } n_0 = \max(n_1, n_2)$$

$$0 \leq f(n) \leq c_1 \cdot g(n) \leq c_1 c_2 \cdot h(n)$$

$$d) 2^{n+1} = O(2^n)$$

$$2^{n+1} = 2 \cdot 2^n$$

$$\forall n \geq 1: 0 \leq 2^{n+1} = 2 \cdot 2^n \leq 2 \cdot 2^n$$

$$2^{n+1} > 0 \xrightarrow{\uparrow} c=2 \text{ (con estas se cumple la definici3n.)}$$

$$n_0 = 1$$